

## TREVICO, Italy

WMO: 162630

Lat: 41.047N

Lon: 15.233E

Elev: 3586

StdP: 12.89

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 21.0       | 24.1 | 9.1                           | 10.0 | 37.1 | 14.3 | 12.9 | 33.2 | 39.6                  | 40.3 | 34.5 | 38.5 | 13.0                  | 0    | 0.607 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 12.5                   | 82.8            | 62.6 | 80.4 | 62.0 | 77.4 | 61.1 | 67.1                | 75.0 | 65.4 | 74.0 | 64.1 | 72.9 | 8.0                  | 280  |

| Dehumidification DP/MCDB and HR |       |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 64.6                            | 104.1 | 70.5 | 62.7 | 97.3 | 69.2 | 61.0 | 91.5 | 68.1 | 33.8          | 74.6 | 32.3 | 74.0 | 31.2 | 73.5 | 75.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |     |      |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|-----|------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |     |      |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max |      |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p) |      |     |
| (4) 30.0          | 26.2 | 23.0 | DB                         | 17.5 | 86.0               | 3.5 | 3.8                                                | 15.0 | 88.7       | 12.9 | 90.9       | 11.0 | 93.0       | 8.5 | 95.7 | (4) |
| (5)               |      |      | WB                         | 16.1 | 70.9               | 3.3 | 2.3                                                | 13.6 | 72.5       | 11.7 | 73.8       | 9.8  | 75.1       | 7.4 | 76.7 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 51.2        | 36.0  | 36.5  | 41.2  | 47.6  | 55.0  | 63.1  | 68.4  | 68.5  | 59.6  | 53.1  | 45.5  | 38.7  | (6)  |  |
|                                                                        | DBStd       | 13.16       | 6.20  | 6.69  | 6.90  | 6.28  | 6.95  | 7.41  | 5.65  | 5.80  | 6.32  | 6.55  | 6.55  | 6.18  | (7)  |  |
|                                                                        | HDD50       | 1805        | 435   | 379   | 281   | 116   | 27    | 3     | 0     | 0     | 6     | 44    | 161   | 353   | (8)  |  |
|                                                                        | HDD65       | 5417        | 900   | 797   | 738   | 524   | 320   | 123   | 29    | 30    | 183   | 371   | 586   | 816   | (9)  |  |
|                                                                        | CDD50       | 2237        | 0     | 2     | 8     | 43    | 182   | 396   | 570   | 574   | 294   | 140   | 26    | 2     | (10) |  |
|                                                                        | CDD65       | 372         | 0     | 0     | 0     | 0     | 10    | 66    | 134   | 139   | 22    | 1     | 0     | 0     | (11) |  |
|                                                                        | CDH74       | 1642        | 0     | 0     | 0     | 1     | 34    | 274   | 646   | 642   | 44    | 1     | 0     | 0     | (12) |  |
|                                                                        | CDH80       | 269         | 0     | 0     | 0     | 0     | 3     | 40    | 110   | 114   | 2     | 0     | 0     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 11.5        | 13.3  | 13.0  | 13.4  | 12.1  | 11.2  | 10.2  | 10.1  | 10.0  | 9.9   | 10.6  | 11.7  | 12.6  | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 24.7        | 2.1   | 1.9   | 2.4   | 2.5   | 1.9   | 1.7   | 1.5   | 1.2   | 2.5   | 2.2   | 3.0   | 2.8   | (15) |  |
|                                                                        | PrecMax     | 39.2        | 6.3   | 3.5   | 6.1   | 5.4   | 5.0   | 3.6   | 4.5   | 4.1   | 7.2   | 5.7   | 9.4   | 5.9   | (16) |  |
|                                                                        | PrecMin     | 18.1        | 0.4   | 0.3   | 0.0   | 0.9   | 0.3   | 0.2   | 0.0   | 0.0   | 0.7   | 0.2   | 0.9   | 0.6   | (17) |  |
|                                                                        | PrecStd     | 4.8         | 1.3   | 0.9   | 1.3   | 1.0   | 1.2   | 0.9   | 1.2   | 0.9   | 1.6   | 1.3   | 1.7   | 1.1   | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 52.1  | 57.1  | 60.0  | 67.9  | 78.4  | 84.2  | 86.4  | 87.8  | 78.7  | 71.2  | 66.0  | 55.1  | (19) |  |
|                                                                        |             | MCWB        | 40.9  | 43.0  | 47.9  | 54.4  | 59.3  | 63.1  | 61.8  | 64.1  | 59.0  | 55.9  | 49.3  | 47.8  | (20) |  |
|                                                                        | 2%          | DB          | 48.5  | 51.6  | 56.3  | 63.3  | 73.0  | 80.5  | 82.7  | 82.7  | 74.8  | 66.5  | 59.2  | 51.7  | (21) |  |
|                                                                        |             | MCWB        | 42.0  | 41.9  | 46.5  | 51.7  | 57.9  | 62.7  | 62.3  | 62.6  | 58.9  | 55.7  | 48.7  | 45.7  | (22) |  |
|                                                                        | 5%          | DB          | 46.2  | 48.0  | 53.3  | 59.3  | 68.6  | 77.0  | 80.4  | 79.7  | 71.6  | 64.2  | 56.7  | 48.4  | (23) |  |
|                                                                        |             | MCWB        | 41.8  | 42.7  | 45.8  | 49.1  | 56.3  | 61.8  | 62.4  | 61.3  | 58.6  | 55.1  | 49.1  | 43.6  | (24) |  |
|                                                                        | 10%         | DB          | 44.2  | 44.8  | 50.3  | 57.0  | 65.9  | 73.8  | 77.2  | 77.2  | 68.4  | 61.6  | 53.7  | 46.4  | (25) |  |
|                                                                        |             | MCWB        | 41.3  | 40.8  | 43.8  | 48.2  | 55.3  | 60.3  | 61.6  | 60.8  | 58.3  | 54.3  | 48.5  | 42.8  | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 46.3  | 48.0  | 51.6  | 57.3  | 62.0  | 68.4  | 69.5  | 69.7  | 64.8  | 61.0  | 55.3  | 49.5  | (27) |  |
|                                                                        |             | MCDB        | 48.8  | 49.8  | 55.8  | 64.5  | 71.9  | 76.8  | 76.0  | 77.0  | 69.8  | 65.0  | 58.9  | 53.1  | (28) |  |
|                                                                        | 2%          | WB          | 44.5  | 45.7  | 48.6  | 53.3  | 59.9  | 65.9  | 67.2  | 66.7  | 63.1  | 58.6  | 53.0  | 47.2  | (29) |  |
|                                                                        |             | MCDB        | 46.2  | 48.6  | 53.8  | 60.4  | 68.7  | 75.1  | 74.9  | 74.8  | 68.7  | 63.1  | 56.3  | 49.9  | (30) |  |
|                                                                        | 5%          | WB          | 42.9  | 43.3  | 46.8  | 51.0  | 58.2  | 63.8  | 65.5  | 65.1  | 61.5  | 57.1  | 51.3  | 45.5  | (31) |  |
|                                                                        |             | MCDB        | 44.8  | 45.8  | 51.4  | 57.1  | 66.9  | 73.4  | 74.0  | 73.9  | 67.5  | 61.4  | 54.7  | 47.6  | (32) |  |
|                                                                        | 10%         | WB          | 41.2  | 41.4  | 45.0  | 49.2  | 56.3  | 62.0  | 64.0  | 63.8  | 59.9  | 55.6  | 49.6  | 43.6  | (33) |  |
|                                                                        |             | MCDB        | 43.2  | 43.6  | 48.8  | 55.1  | 64.0  | 71.6  | 72.7  | 73.0  | 66.1  | 60.0  | 52.7  | 45.8  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 6.5   | 7.7   | 9.3   | 10.6  | 11.8  | 12.5  | 12.9  | 12.5  | 10.2  | 8.6   | 7.4   | 6.7   | (35) |  |
|                                                                        |             | MCDBR       | 7.9   | 10.3  | 12.4  | 13.0  | 13.7  | 14.1  | 14.7  | 13.7  | 12.4  | 10.4  | 9.4   | 8.2   | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 6.3   | 7.9   | 9.4   | 10.2  | 10.3  | 10.0  | 9.4   | 8.9   | 8.6   | 8.0   | 7.6   | 6.6   | (37) |  |
|                                                                        |             | MCWBR       | 6.9   | 8.9   | 11.2  | 11.8  | 13.1  | 13.5  | 13.2  | 12.4  | 10.7  | 9.3   | 8.8   | 7.6   | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.287 | 0.305 | 0.332 | 0.371 | 0.370 | 0.367 | 0.365 | 0.365 | 0.357 | 0.327 | 0.310 | 0.290 | (40) |  |
|                                                                        |             | taud        | 2.552 | 2.490 | 2.411 | 2.292 | 2.336 | 2.368 | 2.380 | 2.386 | 2.423 | 2.517 | 2.530 | 2.552 | (41) |  |
|                                                                        | Ebn at Noon |             | 280   | 290   | 292   | 285   | 287   | 287   | 287   | 284   | 279   | 278   | 268   | 269   | (42) |  |
|                                                                        |             | Edh at Noon | 22    | 28    | 33    | 40    | 39    | 38    | 37    | 36    | 33    | 26    | 23    | 21    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 552   | 794   | 1184  | 1574  | 1931  | 2208  | 2280  | 2022  | 1442  | 1013  | 643   | 504   | (44) |  |
|                                                                        | RadStd      |             | 67    | 84    | 131   | 135   | 140   | 115   | 130   | 137   | 82    | 95    | 62    | 66    | (45) |  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (14 neighbors) | +0.77 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -180  | +259  | +121  |

Nomenclature: See separate page